ITPGA0-33 Project

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Table of contents

1. Project
2. Instructions to Students
3. Section A
4. Terms of AI use for this Assessment
    4.1. Question 1
    4.2. How AI use Affects Marking

Faculty:	Information Systems
Module Code:	ITPGA0-33
Module Name:	Python Programming
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Total Marks:	100 Marks
Submission Week:	Week 6

This module is presented on NQF level 5.

5% will be deducted from the student's project mark for each calendar day the assignment is submitted late, up to a maximum of three calendar days. The penalty will be based on the official campus submission date.

Assignments submitted later than three calendar days after the deadline or not submitted will get 0%. [1]

This is an individual assessment.

This project contributes 30% towards the final mark.

[1] Under no circumstances will assignments be accepted for marking after the assignments of other students have been marked and returned to the students.

Instructions to Students

1. Please ensure that you use the Assignment Template when preparing your submission (where applicable).
2. Please ensure that your answer file (where applicable) is named as follows before submission: **Module Code – Assessment Type – Campus Name – Student Number.**
3. Remember to keep a copy of all submitted assignments.
4. All work must be typed.
5. Please note that you will be evaluated on your writing skills in all your assignments.
6. All work must be submitted through Turnitin. The full originality report will be automatically generated and available for the lecturer to assess. Negative marking will be applied if you are found guilty of plagiarism, poor writing skills, or if you have applied incorrect or insufficient referencing. (See the "instructions to students" book activity before this activity where the application of negative marking is explained.)
7. You are not allowed to offer your work for sale or to purchase the work of other students. This includes the use of professional assignment writers and websites, such as Essay Box. You are also not allowed to make use of artificial intelligence tools, such as ChatGPT, to create content and submit it as your own work. If this should happen, Eduvos reserves the right not to accept future submissions from you.

Section A

Learning Objective

Develop programs in an object-oriented manner

Project Topic

Kasi Fresh Grocers: Inventory & Loyalty Console Application

Scope

AC 2.1 - 2.7

AI Use Requirements for This Assessment

This assessment is classified as AI-1 as defined in the AI Use Category Table below.

AI Use Category Table

For this specific assessment, the following applies:

Code	Category	What This Means	Evidence Required	Disclosure Requirement
AI-1	Limited AI Use	You may use AI only for simple support tasks (e.g. basic language and grammar, clarifying a definition or generating synonyms). The planning, brainstorming, research and final output must still be your own.	You will be required to attach full transcripts of all your AI chats for evidence of your process.	You must complete and submit an **AI Disclosure Appendix** explaining how you used AI. Access the AI Disclosure Appendix Template here.

Terms of AI use for this Assessment

1. What Counts as AI?

AI includes standalone or embedded generative systems that create or materially transform assessed work.

This includes AI features embedded in browsers, search engines, office applications, development environments, coding assistants, data platforms, security tools and engineering software.

The product name is less important than what the feature does.

See What Counts as AI

2. Permitted AI Activities

AI may suggest search terms, but it may not locate, select, summarise or evaluate sources for you.

See Permitted AI Activities

3. Assessed Concepts and General Terminology

You may not use AI to explain an assessed module concept as a substitute for engaging with the approved learning material. Each assessment will distinguish between:

Category	Meaning
Assessed concepts	Module knowledge, methods, technical processes or decisions that you must understand and demonstrate independently
General academic terminology	Terms such as peer review, Boolean operator, introduction or primary source that are not themselves being assessed
Borderline terminology	A term whose status is unclear and must be confirmed with the lecturer before using AI

4. Generally Prohibited Activities

These restrictions apply only where the activity forms part of the published assessment criteria. Assessment-specific instructions take precedence where they expressly permit or provide a tool, dataset, configuration, model, code component or other output. Please refer to the below and apply as appropriate.

See Generally Prohibited Activities

4.1. Discipline-specific Prohibited Activities

Please ensure that you consult the prohibited activity list that is specific to this module's discipline from the four discipline options below.

4.1.1. Software Engineering

4.1.2. Network Security Engineering

4.1.3. Data Science

4.1.4. Robotics

Question 1

100 Marks

Study the scenario and complete the question(s) that follow(s):

> Kasi Fresh Grocers wants a simple console-based application to help branch staff manage stock and customer loyalty points, before a full database-backed system is built later in the qualification. Your task is to design and implement this application using sound object-oriented principles: classes and objects, inheritance, polymorphism, and exception handling with basic data persistence.
>
> You will build up the application in four stages, each demonstrating a specific OOP concept, and finish with a small working console menu that ties everything together.

Suggested Class Design

You are required to follow the suggested class design. Your application must implement the specified inheritance hierarchy, with method overriding used to demonstrate polymorphism.

Class	Key Attributes	Key Methods
Product (base class)	product_id, name, unit_price, quantity_in_stock	restock(qty), sell(qty), display_info()
PerishableProduct(Product)	+ expiry_date, shelf_life_days	is_expired(), display_info() [overridden]
NonPerishableProduct(Product)	+ warranty_months	display_info() [overridden]
Employee (base class)	employee_id, name, branch	calculate_pay(), display_info()
Cashier(Employee)	+ hourly_rate, hours_worked	calculate_pay() [overridden]
Manager(Employee)	+ monthly_salary, bonus	calculate_pay() [overridden]

Class	Key Attributes	Key Methods
Customer	customer_id, name, loyalty_points	add_points(amount), redeem_points(amount)
Inventory	products (list)	add_product(), remove_product(), find_product(), total_stock_value()

#Classes & Objects

1.1 Define a Customer class with an __init__ constructor that sets customer_id, name, and starting loyalty points. (10 marks)

1.2 Define an Inventory class that stores a list of Product objects and provides methods to add, remove, and find products by ID. (10 marks)

1.3 Create at least three Product instances and add them to your Inventory, then print a formatted stock list. (5 marks)

#Inheritance

1.4 Create a Product base class and two concrete subclasses, PerishableProduct and NonPerishableProduct, each adding at least one attribute not present on the base class (e.g. expiry_date, warranty_months). (10 marks)

1.5 Create a second, independent inheritance hierarchy: an Employee base class with Cashier and Manager subclasses, each adding pay-related attributes specific to that role. (10 marks)

1.6 Show that subclass objects can access inherited attributes/methods from their parent class without redefining them. (10 marks)

#Polymorphism

1.7 Give Product and Employee each at least one method that is present on the base class and overridden differently in each subclass (display_info() and calculate_pay() respectively). (10 marks)

1.8 Override display_info() differently in PerishableProduct and NonPerishableProduct, and calculate_pay() differently in Cashier (hourly_rate * hours_worked) and Manager (monthly_salary + bonus). (10 marks)

1.9 Demonstrate polymorphism by looping over a mixed list of Product (or Employee) objects and calling the same method name on each, showing that each object responds according to its own class. (10 marks)

#Exception Handling & Persistence

1.10 Add exception handling around at least three risky operations, e.g. selling more stock than is available (raise/catch a custom or built-in exception), redeeming more loyalty points than a customer has, or loading a data file that does not exist. (5 marks)

1.11 Implement save_to_file()/load_from_file() functionality so that Inventory and Customer data can be saved at the end of a session and reloaded at the start of the next one. (5 marks)

1.12 Tie everything together in a simple text-menu loop (e.g. 1. View stock, 2. Sell product, 3. Manage loyalty points, 4. Save & exit) that uses the classes built in questions 1.1 to 1.9. (5 marks)

Submit a single, well-commented python file (or a small package of modules with a clear entry point) implementing all classes and the console menu.

End of Question 1

How AI use Affects Marking

Finding	Outcome
No AI used and accurately declared	Your lecturer will mark according to the marking guide.
Permitted AI use accurately declared	Your lecturer will mark according to the marking guide.
AI-generated in identifiable assessed component	Your lecturer will mark authentic work and will not award marks for the affected component that is identified to be AI-generated.
AI-generated in dependent criteria may be affected	Your lecturer will evaluate each criterion in isolation and will not award marks for affected criteria that are AI-generated.
Technically correct work that raises authenticity concerns*	Your lecturer will complete the review before finalising the mark, and will not award marks for affected criteria that are AI-generated.
Serious authenticity concerns exist*	Your lecturer may request a formal discussion with you, and depending on the outcome may escalate a case for formal academic-integrity hearing.

*See General Prohibited Activity list under Terms of AI use for this Assessment